

AI-POWERED FAKE PRODUCT REVIEW DETECTION USING NLP

Classifying customer reviews as **Likely Fake** or **Likely Genuine** with TF-IDF + Multinomial Naive Bayes

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01 ABSTRACT

Online product reviews strongly influence customer purchasing decisions, but fake and computer-generated reviews can mislead consumers and reduce trust in e-commerce platforms. This project develops an NLP-based machine learning system to classify product reviews as **Likely Fake** or **Likely Genuine**. The system preprocesses review text, extracts **TF-IDF** features, and applies a **Multinomial Naive Bayes** classifier. A Streamlit-based application provides real-time review analysis and supporting linguistic insights.

02 PROBLEM STATEMENT

- Online shoppers increasingly rely on customer reviews before purchasing.
- Fake or computer-generated reviews can artificially influence customer decisions.
- Highly promotional, repetitive, or suspicious review language can reduce trust.
- Manual identification of fake reviews is difficult and time-consuming.
- A simple AI-assisted review analysis system can provide additional decision support.

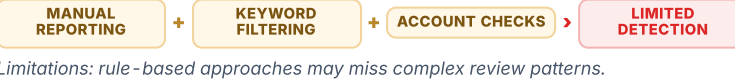


03 OBJECTIVES

- Develop an NLP-based fake product review classification system.
- Preprocess customer review text consistently for training and prediction.
- Extract meaningful textual features using TF-IDF.
- Classify reviews using Multinomial Naive Bayes.
- Provide a user-friendly Streamlit application for real-time review analysis.

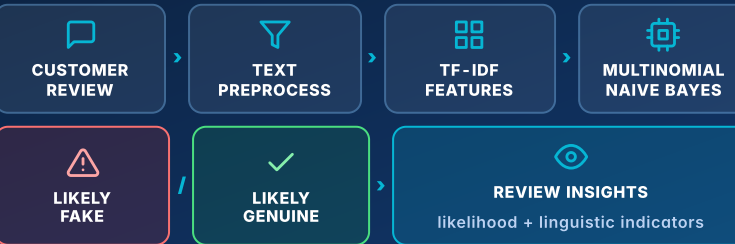
04 EXISTING SYSTEM

- Manual user reporting of suspicious reviews.
- Basic keyword or blacklist filtering.
- Account-level frequency checks.
- Limited understanding of review language and context.
- Difficulty identifying sophisticated or computer-generated reviews.



05 PROPOSED SYSTEM

An NLP and machine-learning based system that analyzes the language patterns of customer reviews and classifies them as **Likely Fake** or **Likely Genuine**.



LANGUAGE-AWARE DECISION SUPPORT

Learns review-language patterns from labeled data instead of fixed rules — a prediction supports, not replaces, human judgement.

06 METHODOLOGY

STEP 1 DATASET

40,432
labeled product reviews
CG (fake) / OR (genuine)

STEP 2 TEXT PRE-PROCESSING

- Lowercase normalization
- URL removal
- Punctuation cleaning
- Whitespace normalization
- Stopword filtering

STEP 3 TRAIN / TEST SPLIT

- 80% Training
- 20% Testing
- Stratified split

STEP 4 TF-IDF

25,000
TF-IDF features
Unigram + Bigram

STEP 5 CLASSIFICATION

- Multinomial Naive Bayes
- Fake vs Genuine

STEP 6 EVALUATION

- Accuracy
- Precision · Recall
- F1-score
- Confusion Matrix

STEP 7 APPLICATION

- Streamlit web app
- Real-time review prediction

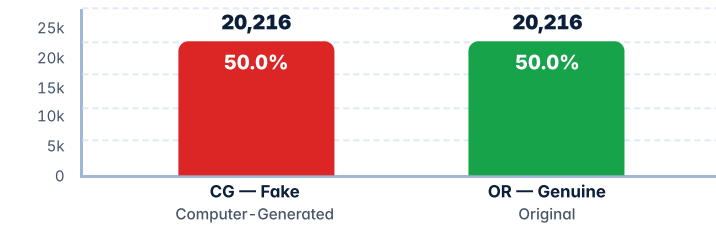
END-TO-END WORKFLOW

07 NLP PIPELINE



09 DATASET

FAKE REVIEWS DATASET
40,432 reviews
CLASS DISTRIBUTION

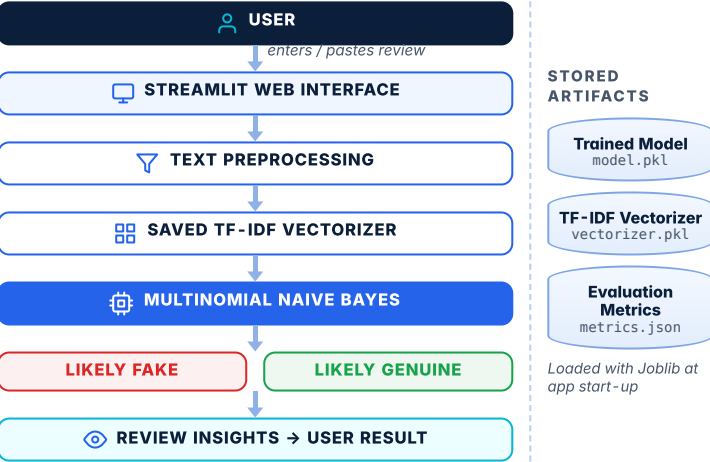


0 Missing values | 12 Exact duplicate rows | 20 Duplicate review texts | 10 Product categories | 1.0-5.0 Rating range

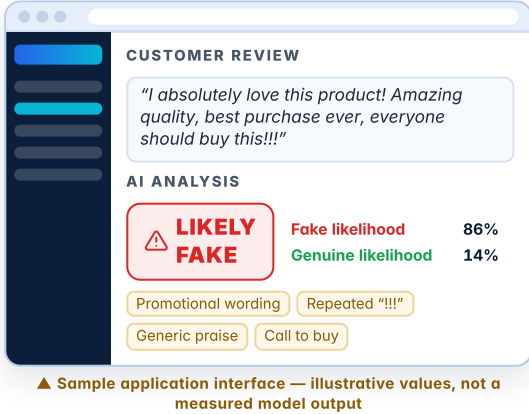
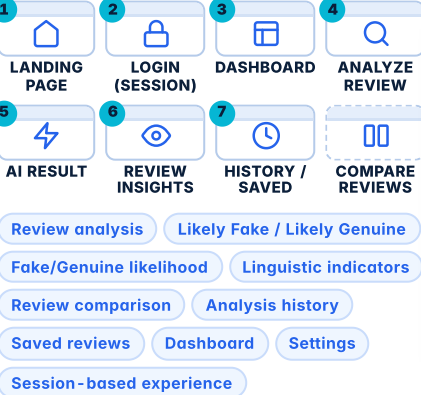
10 TECHNOLOGY STACK



08 SYSTEM ARCHITECTURE



11 IMPLEMENTATION & APPLICATION OUTPUT



▲ Sample application interface — illustrative values, not a measured model output

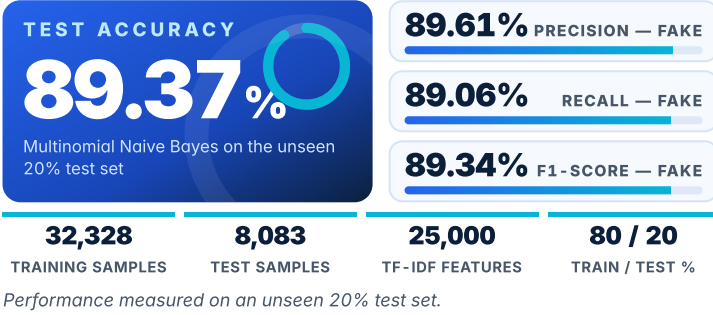
14 KEY FINDINGS

- Balanced dataset: **20,216 Fake** and **20,216 Genuine** reviews.
- TF-IDF unigram + bigram features capture useful review-language patterns.
- Multinomial Naive Bayes achieved **89.37%** accuracy on the unseen test set.
- Fake-review precision and recall were both $\approx$ **89%**.
- Real-time review classification through Streamlit.
- Linguistic indicators are supporting information, not definitive proof of authenticity.

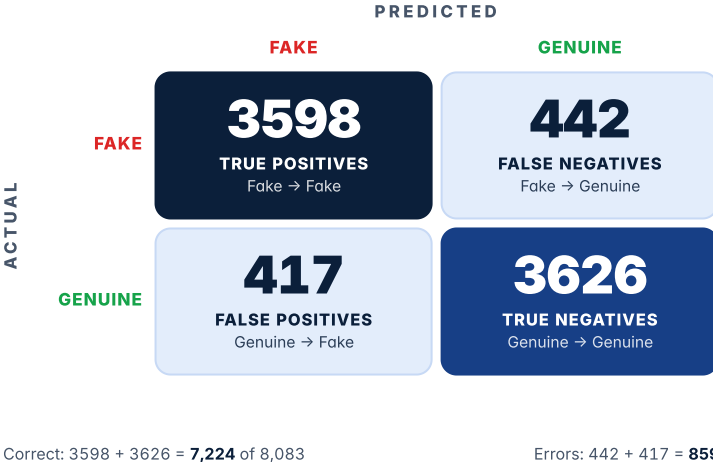
CONCLUSION

The project demonstrates an effective NLP-based approach for identifying potentially fake product reviews. By combining consistent text preprocessing, TF-IDF feature extraction, and Multinomial Naive Bayes, the system achieved **89.37% test accuracy** on the evaluated dataset. The Streamlit application translates the trained model into an accessible review-analysis tool that provides Likely Fake/Likely Genuine predictions and supporting linguistic insights.

12 MODEL PERFORMANCE



13 CONFUSION MATRIX



16 FUTURE SCOPE

- Analyze complete product pages / product-level review collections.
- Larger, more diverse real-world review datasets.
- Compare additional ML and deep-learning approaches.
- Explore transformer models such as BERT / RoBERTa.
- Large-scale e-commerce review moderation and integration.
- Improve explainability and multilingual analysis.

18 LIMITATIONS

- Classifier analyzes review text; it does not independently verify the reviewer.
- A Likely Fake prediction is not proof that a review is fraudulent.
- Performance depends on the training dataset and its language patterns.
- Current prototype focuses primarily on English review text.
- Product-level URL analysis is future scope, not a current core capability.

17 REFERENCES

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